

STEPHANIE SCHAUPP

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EDUCATION

University of California, Berkeley
B.S. Energy Engineering

Berkeley, CA
Expected graduation: **May 2027**

PROFESSIONAL EXPERIENCE

Boles Research Lab • Power Electronics Research Assistant *Aug 2026 – Present • Berkeley, CA*

- Extending an automated testbed for nonlinear mechanical loss in piezoelectric resonators, expanding characterization beyond three PZT materials (APC 841/844/880) to new material types and geometries, and refining empirical Steinmetz loss models
- Optimizing the C2000 sweep algorithm to find resonance and peak-efficiency points faster, adapting for fragile materials

Rivian & Volkswagen Group Technologies • Hardware Validation Intern *May 2026 – Present • Palo Alto, CA*

- Improved RF ECU board testability from 45.6% to 56.7% in Valor by identifying test-critical nets, potential test access points, and implementing design changes within DFM constraints, bringing the board into compliance with production test specifications
- Authored a cross-organizational DFT guidelines document aligning test point placement standards across Rivian, Volkswagen, and all 4 CMs, covering edge clearance, mechanical mounting, component proximity, sizing, and layout constraints
- Built a cross-CM failure analytics pipeline, automating ICT/FVT/EOL test data ingestion, failure categorization, and RCA, and integrating with the company-native FA tool, reducing FA debug cycles across contract manufacturers

Siemens EDA • PCB Design Engineer Intern *Jan 2026 - May 2026 • Remote, CA*

- Designed and routed a 4-layer Cyclone IV FPGA-based flight controller PCB in Xpedition Designer, capturing the full schematic across RF transceiver, IMU, clock generator, JTAG, 4 motor drivers, and a 4-rail DC-DC power network with 100+ I/O pins across 8 banks, then completing layout optimized for range vs. board area and delivering through DRC and BOM
- Validated board signal and power integrity by investigating static overshoot, ringing, crosstalk, DC droop, and decoupling on critical nets, designing targeted termination, trace spacing, and length, and power plane changes to meet specs at minimal cost

Pilawa Research Group • Power Electronics Research Assistant *Sep 2025 – May 2026 • Berkeley, CA*

- Overhauled a legacy 500V lab safety system, reordering and bringing up all hardware from scratch, validating original schematics, and redesigning components to support 1kV+ operation, doubling the original specifications for FCML testing

Zoox (Amazon) • Hardware Manufacturing TPM Intern *May 2025 – Aug 2025 • Foster City, CA*

- Designed data collection methods and implemented quality and production KPI tracking, aligning mechanical and electrical design intent with production constraints across prototype, early, and full-scale production, increasing FPY by 12%
- Collaborated with the Prototype Engineering Team to automate data collection and analysis of 1000+ hardware quality inspections, enabling trend analysis, root-cause tracking, and issue ownership for electromechanical failures across subsystems

CalSol • Hardware Engineer *Jan 2024 – Jan 2025 • Berkeley, CA*

- Assembled and validated a 17-module, 1 kWh lithium-ion pack for solar car racing, verifying cell-level performance, thermal behavior, pack-level energy output against design targets, and integration with the solar electrical system
- Designed mechanical enclosures and electrical interfaces in SolidWorks, optimizing vibration reduction for structural integrity and minimizing contact resistance across high-current paths to improve battery efficiency
- Led a DFM-focused PCB redesign in KiCAD, eliminating busbars, increasing Cu weight to 2 oz, isolating sensitive components, reducing reworks by 70%, and decreasing build time by 1.5 weeks out of a 5-week manufacturing design cycle

PROJECTS

MPPT Synchronous Buck *Jan 2026 – May 2026*

- Designed a 20V–12V synchronous buck converter for PV battery charging in Altium Designer, selecting a 50 kHz switching frequency and an RM10 core to minimize inductor I_{RMS} , switching loss, and switch stress while optimizing capacitor selection for current sharing and ESR loss reduction, balancing trade-offs between efficiency, power density, and ripple specifications
- Validated stable regulation and safe switching across a 16V–24V input range and 50–100W power sweep, achieving 97.51% peak efficiency, 5% output voltage ripple, 96.2% dynamic MPPT tracking efficiency, and safe thermal performance

SKILLS

EE & Hardware: Oscilloscope, DC Power Supply, Soldering/Rework, Battery Pack Assembly

EDA & Design Tools: Altium Designer, Xpedition Designer/Layout, HyperLynx, Valor, KiCad, LTspice, PLECS, SolidWorks

Manufacturing & Validation: DFM, DFT, Signal Integrity, Power Integrity, CPK, FAI

Embedded & Programming: C (TI C2000, CCS, ESP32), Python, MATLAB

Other: Korean (Professional Proficiency), Jira, Confluence, Looker, Excel